# TR-Core: Autonomous Symbolic Induction of Compositional Rules via Apriori Search with Binomial Filtering

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**Abstract**

We present TR-Core, a developmental cognitive architecture that autonomously discovers compositional symbolic rules from raw interaction data. Unlike neural approaches requiring pretraining, TR-Core's behavioral program learns via seven development operations (CREATE, MODIFY, GENERALIZE, SPECIALIZE, COMPOSE, SPLIT, DELETE) over a fixed substrate. The core contribution is an Apriori-style induction engine with binomial statistical filtering that discovers rare compositional rules (conjunctions of 3–4 literals) and generalizes to structurally held-out contexts.

In Phase 3 (rare compositional rules, 8 attributes, 4608 contexts), TR-Core achieves balanced accuracy 0.78–0.80, significantly outperforming kNN (0.70), Random Forest (0.66), and Logistic Regression (0.20). Ablation studies isolate COMPOSE as the causal mechanism: disabling it collapses performance to chance (0.20). In Phase 4 (Structural-OOD generalization), TR-Core generalizes a learned pair rule ($calm \wedge guard \rightarrow approach$) to a never-observed triple ($calm \wedge guard \wedge night \rightarrow approach$) with 96% held-out accuracy, while decision lists (0.91) and decision trees (0.98) overfit to training conjunctions.

We provide formal complexity analysis: worst-case time $\mathcal{O}(L^K)$, expected time $\mathcal{O}((1+\beta L)^K)$ due to anti-monotonicity pruning, space $\mathcal{O}(N + L)$, with exact binomial test controlling per-candidate Type I error at $\alpha = 10^{-3}$. The combinatorial explosion is tamed by a coverage floor inducing a phase transition to subcritical branching.

## 1 Introduction

Modern cognitive architectures [6, 8, 3] typically rely on hand-coded knowledge or neural pretraining. TR-Core investigates a different hypothesis: *can a tabula rasa system autonomously induce compositional symbolic rules from uniform-random interaction, with statistical guarantees?*

The behavioral program framework [4] maintains a layer of changeable rules over a fixed substrate. Rules are condition-action pairs with evidence counters; development operations modify the rule base based on observed outcomes. The key innovation is the Apriori induction engine (`BehavioralProgramV3`) that searches conjunctions up to arity $K = 4$ over $L \approx 20$ literals, using coverage-based pruning and a binomial filter ($p \leq 10^{-3}$, precision $\geq 0.95$).

This paper makes three contributions:

1. **Formal analysis** of the Apriori induction engine: complexity bounds, statistical guarantees, and recovery theorems (Section 4, Appendix A).
2. **Empirical validation** on two controlled protocols: Phase 3 (rare compositional rules) and Phase 4 (Structural-OOD generalization), with 24-seed replications and mechanistic controls.
3. **Professional infrastructure**: Gymnasium environments, Hydra configuration, DVC pipeline, and publication-ready figure generation for reproducibility.

## 2 Related Work

**Symbolic induction.** Classic ILP systems (FOIL [11], Progol [10]) learn Horn clauses from examples but assume oracle labels. Our setting is harder: the agent only observes success/failure for its *chosen* action (bandit feedback), not the oracle action.

**Neural-symbolic integration.** Neural Theorem Provers [12], DeepProbLog [9], and GNNs [2] combine neural perception with symbolic reasoning but require differentiable relaxations or pre-training. TR-Core operates with purely symbolic, non-differentiable operations.

**Developmental robotics.** Architectures like DAC [14] and IM-CLeVeR [1] model developmental stages but lack formal induction guarantees. Our binomial filter provides exact statistical control.

**Compositional generalization.** SCAN [7], COGS [5], and gSCAN [13] test compositional generalization in language navigation. Our Structural-OOD protocol (Phase 4) isolates a specific mechanism: generalization from a learned pair to a held-out triple with zero training support.

## 3 Architecture

### 3.1 Behavioral Program

The behavioral program maintains a rule base $\mathcal{R}$ where each rule $r = (C, a, s, e)$ has condition $C$ (conjunction of literals), action $a$, status $s \in \{\text{LATENT}, \text{ACTIVE}, \text{RETIRED}, \text{DELETED}\}$, and evidence $e = (n_s, n_f)$. Seven operations modify $\mathcal{R}$:

- `CREATE`: new LATENT rule from successful interaction
- `GENERALIZE`: replace constants with variables ($\in$ set)
- `SPECIALIZE`: add discriminant literal to split a rule
- `SPLIT`: partition rule along an attribute
- `COMPOSE`: conjoin two rules into a subprogram
- `MODIFY`: change condition or action
- `DELETE`: retire rule as last resort

### 3.2 Apriori Induction Engine

The `develop()` method (Algorithm **??**) runs periodically on the agent's unbiased explore trials $\mathcal{D}_a$ for each action $a$:

**Algorithm 1: Apriori Induction for One Action**

**Input:** $\mathcal{D}_a$, literals $\mathcal{L}$, max arity $K$, thresholds $(c_{\min}, h_{\min}, \theta, \alpha)$ $S_1 \leftarrow \{\{l\} : l \in \mathcal{L}\}$ **for** $k = 1$ to $K$ $S_k^{\text{supp}} \leftarrow \emptyset$ **for** $C \in S_k$ Compute $\text{cov}_a(C), \text{hits}_a(C)$ via bitmask intersection **if** $\text{cov}_a(C) \geq c_{\min}$ $S_k^{\text{supp}} \leftarrow S_k^{\text{supp}} \cup \{C\}$ **if** $\text{hits}_a(C) \geq h_{\min} \wedge \text{prec}_a(C) \geq \theta \wedge p\text{-value}(C) \leq \alpha$ Materialize rule $C \to a$ via COMPOSE (if $k \geq 2$) or GENERALIZE **end if end if end for if** $S_k^{\text{supp}} = \emptyset$ **break end if** $S_{k+1} \leftarrow \text{GenCandidates}(S_k^{\text{supp}})$ **end for**

Anti-monotonicity of coverage ($\text{cov}(C') \leq \text{cov}(C)$ for $C' \supset C$) enables Apriori pruning. The binomial filter computes exact $p$-value against base rate $p_a$: $p\text{-value}(C) = \mathbb{P}_{\text{Binom}(n,p_a)}(X \geq h)$.

# 4 Theoretical Analysis

## 4.1 Complexity Bounds

**Theorem 1** (Worst-Case Time). *The worst-case number of candidates evaluated is* $\sum_{k=1}^{K} \binom{L}{k} = \mathcal{O}(L^K)$. *With bitmask evaluation (*$\mathcal{O}(1)$ *per candidate after* $\mathcal{O}(NL)$ *preprocessing), total time is* $\mathcal{O}(NL + L^K)$.

**Theorem 2** (Expected Time Under Pruning). *Let* $p_k = \mathbb{P}(\text{cov}(C) \geq c_{\min})$ *for random* $k$*-conjunction. If* $p_k \leq \beta^k$ *for* $\beta < 1$, *expected candidates is* $\mathcal{O}((1 + \beta L)^K)$.

In TR-Core ($L \approx 20$, $K = 4$, $c_{\min} = 4$, $N_a \approx 600$), $p_k$ drops sharply; observed candidates $\sim 10^5$ with $\sim 100$ rules materialized.

**Theorem 3** (Space Complexity). $\mathcal{O}(N + L + RK)$ *where* $R$ *is materialized rules. Bitmasks* $\mathcal{O}(N)$, *literal index* $\mathcal{O}(L)$, *rules* $\mathcal{O}(RK)$.

## 4.2 Statistical Guarantees

**Theorem 4** (Per-Candidate Type I Error). *Under null (true precision =* $p_a$*), probability of materializing a spurious rule* $\leq \alpha = 10^{-3}$.

**Theorem 5** (Detection Power). *For true precision* $p > p_a$, *coverage* $n$, *power is* $\mathbb{P}_{\text{Binom}(n,p)}(X \geq h_{crit})$ *where* $h_{crit}$ *is the critical value at level* $\alpha$. *The filter requires high precision on sufficient coverage by design.*

## 4.3 Recovery Guarantees

**Theorem 6** (Ground-Truth Recovery). *If a ground-truth rule* $C^* \to a$ *has* $\text{cov}_a(C^*) \geq c_{\min}$, $\text{prec}_a(C^*) \geq \theta + \delta$, *and p-value*$(C^*) \leq \alpha/2$ *with high probability, then* $C^*$ *is materialized.*

Phase 3 validates this: 4/6 oracle rules recovered in 100% of seeds; 2/6 in 50–83% (missed due to priority overlap, not coverage).

# 5 Experiments

## 5.1 Phase 3: Rare Compositional Rules

**Environment V3.** 8 attributes (role, friendliness, status, mood, context, season, location, time), 4608 contexts, 5 actions. Oracle contains 4 rare compositional rules (arity 3–4) whose single literals are individually ambiguous (precision 0.16–0.68); only conjunctions disambiguate.

**Protocol.** Train=2500, known=200, test=600 (3 splits). Uniform-random explore actions. Metrics: balanced accuracy (5 classes imbalanced), rule recovery, behavioral FDR.

**Baselines.** kNN, Decision Tree, Random Forest, Logistic Regression, Decision List (arity-1), Multi-Decision List (arity-4, greedy), CompoLift (Apriori search with lift scoring), plus mechanistic controls: `null_bp` (randomized outcomes), `ab_no_compose` (COMPOSE disabled).

**Results (Table 1).** TR-Core (full_bp) achieves 0.784 balanced accuracy (main), 0.803 (replication). COMPOSE is necessary: ablations collapse to 0.20 (chance). TR-Core beats kNN ($p < 10^{-4}$, $d_z = 1.39$) and RF ($d_z = 1.64$). Multi-Decision List (0.813) matches TR-Core in main ($p = 0.10$) but not replication ($p = 0.049$), confirming compositional *search* is the key, not specific scoring.

Table 1: Phase 3 Balanced Accuracy (N=24 seeds)

| Model | Main | Replication |
|---|---|---|
| **full_bp** | **0.784** | **0.803** |
| ValueMultiDecisionList | 0.813 | 0.766 |
| ValueDecisionTree | 0.752 | 0.763 |
| ValuekNN | 0.702 | 0.710 |
| ValueRandomForest | 0.659 | 0.671 |
| ValueCompoLift | 0.681 | 0.685 |
| ValueLogisticRegression | 0.200 | 0.200 |
| ValueDecisionList (arity-1) | 0.200 | 0.200 |
| null_bp | 0.200 | 0.202 |
| ab_no_compose | 0.200 | 0.200 |
| chance | 0.199 | 0.205 |
| ceiling | 1.000 | 1.000 |

**Rule Recovery.** 6 oracle compositional rules targeted. Exact/subset recovery rate: 0.68 main, 0.66 replication. Behavioral FDR (policy-level precision): 0.36 (36% of ACTIVE compositional rules cause at least one error). Winner accuracy on full population: 0.916.

### 5.2 Phase 4: Structural-OOD Generalization

**Environment V4.** 6 attributes, 864 contexts, 4 actions. Sub-protocol: held-out triple `calm` ∧ `guard`∧`night` → `approach` has *zero* training support. The covering pair `calm`∧`guard` → `approach` is a training rule. Only compositional generalization from the pair succeeds; memorizers (kNN) reach only seen sub-patterns.

**Results (Table 2).** TR-Core held-out accuracy: 0.958 (main), 1.000 (replication). Pair rule recovered 24/24 seeds. Decision Tree (0.977) and Multi-Decision List (0.910) match on held-out but differ on balanced accuracy (TR-Core 0.693 vs Tree 0.752). kNN fails (0.253). Ablations confirm COMPOSE as mechanism: `ab_no_compose` and `null_bp` at 0.000 held-out.

Table 2: Phase 4 Held-out Accuracy (Structural-OOD)

| Model | Main (heldout) | Replication (heldout) |
|---|---|---|
| **full_bp** | **0.958** | **1.000** |
| ValueDecisionTree | 0.977 | 0.985 |
| ValueMultiDecisionList | 0.910 | 0.995 |
| ValueRandomForest | 0.844 | 0.935 |
| ValuekNN | 0.253 | 0.311 |
| ab_no_compose | 0.000 | 0.000 |
| null_bp | 0.000 | 0.000 |
| chance | 0.251 | 0.236 |
| ceiling | 1.000 | 1.000 |

## 6 Discussion

**Why Apriori + Binomial works.** The coverage floor $c_{\min} = 4$ induces a phase transition: effective branching factor $b_{\text{eff}}(k) < 1$ for $k \geq 3$, making search tractable. The binomial filter eliminates low-coverage spurious rules that would otherwise proliferate (without filter: 206 rules/seed; with filter: 100).

**Limitations.** (1) Binomial filter weakens for rules with few samples in actively-chosen actions (e.g., `warm/stranger/spring/dawn` 13/24 seeds). (2) Arity $\geq 5$ and "$\neq$" literals not yet supported. (3) Priority overlap in oracle causes irreducible FDR ( 0.36).

**Future work.** Hierarchical planner over learned rules; transfer to continuous/POMDP environments; neural-symbolic baselines (GNN, NTP).

## 7 Limitations and Boundaries

**Discreteness.** In v0.3, input data requires quantization (discretization of continuous features). The core expects structured predicates and requires an external encoder for raw signals (pixels, audio).

**Perceptual dependency.** The core operates on structured predicates. Working with raw signals requires an external encoder (perception layer), as planned in Alan Phase 1.

**Asymptotic complexity.** Without pruning: $O(2^N)$ worst-case. With pruning enabled: $O(N \log N)$ under the assumption of sparse data and subcritical branching (coverage floor $c_{\min} = 4$). The actual complexity depends on data structure and attribute correlation.

**Scalability.** Stress benchmark results (Table 3) show latency and memory scaling with attribute count $N$.

Table 3: Stress Benchmark: Attribute Scalability (N attrs, 1000 train samples, max arity 2)

| $N$ (attrs) | Latency (ms) | Peak RAM (MB) | OOD Acc |
|---|---|---|---|
| 8 | 119.5 | 0.16 | 1.0000 |
| 16 | 688.8 | 0.55 | 1.0000 |
| 32 | 3303.7 | 2.32 | 1.0000 |

**Binomial filter sensitivity.** The filter weakens for rules with few samples in actively-chosen actions (e.g., `warm/stranger/spring/dawn` recovered 13/24 seeds).

**Arity limit.** Maximum arity $K = 4$; arity $\geq 5$ and $\neq$ literals not yet supported.

## 8 Conclusion

TR-Core demonstrates that a tabula rasa architecture with Apriori induction and binomial filtering can autonomously discover compositional symbolic rules and generalize to structurally held-out contexts. The formal analysis establishes tractability (subcritical branching) and statistical control (exact binomial test). Empirical results on two controlled protocols with mechanistic controls isolate COMPOSE as the causal mechanism for compositional generalization.

## References


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# A Formal Complexity Analysis

# B Problem Setup and Notation

## B.1 Attribute Domain

Let $\mathcal{A} = \{A_1, \ldots, A_m\}$ be a set of $m$ categorical attributes. Each attribute $A_i$ has a finite domain $\mathrm{dom}(A_i)$ of size $d_i = |\mathrm{dom}(A_i)|$. A *literal* is a pair $(A_i, v)$ with $v \in \mathrm{dom}(A_i)$, denoted $A_i = v$. The set of all possible literals is

$$\mathcal{L} = \bigcup_{i=1}^{m} \{A_i\} \times \mathrm{dom}(A_i), \qquad |\mathcal{L}| = \sum_{i=1}^{m} d_i =: L.$$

In TR-Core v3/v4: $m = 5$ or 8, $d_i \in \{2, 3, 4\}$, so $L \in [15, 25]$.

## B.2 Conjunctions and Arity

A *conjunction* (candidate condition) of arity $k$ is a set $C \subseteq \mathcal{L}$ with $|C| = k$ and at most one literal per attribute. The set of all valid conjunctions of arity $\leq K$ is

$$\mathcal{C}_{\leq K} = \bigcup_{k=1}^{K} \{C \subseteq \mathcal{L} : |C| = k,\ \forall A_i, |C \cap (\{A_i\} \times \mathrm{dom}(A_i))| \leq 1\}.$$

Total candidates:

$$|\mathcal{C}_{\leq K}| = \sum_{k=1}^{K} \sum_{1 \leq i_1 < \cdots < i_k \leq m} \prod_{j=1}^{k} d_{i_j}.$$

For uniform domains $d_i = d$, this is $\sum_{k=1}^{K} \binom{m}{k} d^k \leq (1 + d)^m - 1$.

## B.3 Data and Coverage

The agent collects *explore trials* $\mathcal{D} = \{(x^{(t)}, a^{(t)}, y^{(t)})\}_{t=1}^{N}$ where $x^{(t)} \in \mathcal{X} = \prod_i \mathrm{dom}(A_i)$ is a context, $a^{(t)} \in \mathcal{Y}$ is an action chosen uniformly at random (exploration), and $y^{(t)} \in \{0, 1\}$ is success. For a fixed action $a$, define the filtered dataset

$$\mathcal{D}_a = \{(x, y) \in \mathcal{D} : a^{(t)} = a\}, \qquad N_a = |\mathcal{D}_a|.$$

For a conjunction $C$, its *coverage* and *hits* on $\mathcal{D}_a$ are

$$\mathrm{cov}_a(C) = |\{(x, y) \in \mathcal{D}_a : x \models C\}|, \qquad \mathrm{hits}_a(C) = |\{(x, y) \in \mathcal{D}_a : x \models C \wedge y = 1\}|.$$

Precision is $\mathrm{prec}_a(C) = \mathrm{hits}_a(C)/\mathrm{cov}_a(C)$ (undefined if $\mathrm{cov}_a(C) = 0$).

### B.4 Induction Criteria

A candidate $C$ for action $a$ is *materialized* as a rule iff:

1. $\text{cov}_a(C) \geq c_{\min}$ (coverage floor)
2. $\text{hits}_a(C) \geq h_{\min}$ (minimum successes)
3. $\text{prec}_a(C) \geq \theta$ (precision threshold)
4. $p$-value $= \mathbb{P}_{\text{Binom}(N_a, p_a)}(\text{hits} \geq \text{hits}_a(C)) \leq \alpha$ (binomial filter)

where $p_a = \text{hits}_a(\emptyset)/N_a$ is the base success rate for action $a$. Default parameters: $c_{\min} = 4$, $h_{\min} = 5$, $\theta = 0.95$, $\alpha = 10^{-3}$, $K = 4$.

## C Algorithm: Apriori Induction with Binomial Filter

**Algorithm 1: Apriori Induction for One Action**

**Input:** $\mathcal{D}_a$, $L$ literals, $K$, thresholds $(c_{\min}, h_{\min}, \theta, \alpha)$ $S_1 \leftarrow \{\{l\} : l \in \mathcal{L}\}$ **for** $k = 1$ to $K$ $S_k^{\text{supported}} \leftarrow \emptyset$ **for** $C \in S_k$ Compute $\text{cov}_a(C), \text{hits}_a(C)$ via bitmask intersection **if** $\text{cov}_a(C) \geq c_{\min}$ $S_k^{\text{supported}} \leftarrow S_k^{\text{supported}} \cup \{C\}$ **if** $\text{hits}_a(C) \geq h_{\min} \wedge \text{prec}_a(C) \geq \theta \wedge p\text{-value}(C) \leq \alpha$ Materialize rule $C \to a$ **end if end if end for if** $S_k^{\text{supported}} = \emptyset$ **break end if** $S_{k+1} \leftarrow \text{GenCandidates}(S_k^{\text{supported}})$ **end for**

### C.1 Candidate Generation (Apriori Join)

GenCandidates$(S)$ produces all $(k+1)$-literal sets whose every $k$-subset is in $S$:

$$\text{GenCandidates}(S) = \{C \cup \{l\} : C \in S,\ l \in \mathcal{L} \setminus C,\ \forall C' \subset C \cup \{l\}, |C'| = k \implies C' \in S\}.$$

In the implementation (line 154-169 of `program_v3.py`), this is done by extending each supported $C$ with literals not in $C$, then sorting for determinism.

## D Complexity Analysis

### D.1 Worst-Case Time Complexity

**Theorem 7** (Worst-Case Candidate Enumeration)**.** *The worst-case number of candidates evaluated across all arities* $1 \dots K$ *is*

$$T_{cand} = \sum_{k=1}^{K} \binom{L}{k} = \mathcal{O}(L^K).$$

*Proof.* Without pruning, every subset of $\mathcal{L}$ of size $\leq K$ is a candidate. The sum of binomial coefficients is bounded by $(eL/K)^K$ for $K \ll L$, i.e., $\mathcal{O}(L^K)$. □

**Theorem 8** (Worst-Case Coverage Computation)**.** *For a single candidate* $C$*, computing* $\text{cov}_a(C)$ *and* $\text{hits}_a(C)$ *over* $N_a$ *trials takes* $\mathcal{O}(N_a)$ *time naively, or* $\mathcal{O}(1)$ *with precomputed bitmasks (as implemented). With bitmasks:*

$$T_{eval} = \mathcal{O}(N_a \cdot L) \text{ preprocessing} + \mathcal{O}(T_{cand}) \text{ evaluation.}$$

*Proof.* Preprocessing: tokenize each trial into a bitmask of length $L$ ($\mathcal{O}(N_a L)$). Evaluation: bitwise AND and popcount are $\mathcal{O}(1)$ on 32/64-bit words (here $L \leq 32$ fits in `uint32`). □

**Corollary 1** (Overall Worst-Case Time)**.**

$$T_{worst} = \mathcal{O}(N_a L + L^K).$$

*For TR-Core: $N_a \sim 2500$, $L \sim 20$, $K = 4 \implies L^K = 160{,}000$ candidates, feasible.*

### D.2 Expected Complexity Under Anti-Monotonicity

**Definition 1** (Anti-Monotonicity of Coverage)**.** *Coverage is anti-monotone: if $C \subseteq C'$ then $\mathrm{cov}_a(C') \leq \mathrm{cov}_a(C)$.*

**Lemma 1** (Apriori Pruning Correctness)**.** *If $\mathrm{cov}_a(C) < c_{\min}$, then for all $C' \supset C$, $\mathrm{cov}_a(C') < c_{\min}$.*

*Proof.* Direct from anti-monotonicity: $C \subseteq C' \implies \mathrm{cov}_a(C') \leq \mathrm{cov}_a(C) < c_{\min}$. □

**Theorem 9** (Expected Candidates Under Pruning)**.** *Let $p_k = \mathbb{P}(\mathrm{cov}_a(C) \geq c_{\min})$ for a random $k$-literal conjunction. Assume $p_k$ decays exponentially: $p_k \leq \beta^k$ for some $\beta < 1$. Then expected candidates:*

$$\mathbb{E}[T_{cand}] = \sum_{k=1}^{K} \binom{L}{k} p_{k-1} \leq \sum_{k=1}^{K} \binom{L}{k} \beta^{k-1} = \mathcal{O}((1+\beta L)^K).$$

*Proof.* A $k$-candidate is generated only if its $(k-1)$-subsets are supported. By union bound, probability a random $k$-candidate is generated $\leq \beta^{k-1}$. Sum over all $\binom{L}{k}$ candidates. □

**Remark 1.** *In practice, $c_{\min} = 4$ and sparse data ($N_a \sim 600$) make $p_k$ drop sharply. Phase 3 logs show $\sim 100$ rules materialized per seed from $\sim 10^5$ initial candidates, implying effective branching factor $\ll L$.*

### D.3 Space Complexity

**Theorem 10** (Space Complexity)**.**

$$S = \mathcal{O}(N_a + L + R \cdot K)$$

*where $R$ is the number of materialized rules. Bitmasks: $\mathcal{O}(N_a)$; literal index: $\mathcal{O}(L)$; rules: $\mathcal{O}(RK)$.*

In TR-Core: $R \sim 100$, $K = 4$, $N_a \sim 2500 \implies$ negligible memory.

## E Statistical Guarantees of the Binomial Filter

### E.1 Filter Definition

For candidate $C$ with $\mathrm{cov}_a(C) = n$, $\mathrm{hits}_a(C) = h$, the filter computes

$$p\text{-value}(C) = \sum_{j=h}^{n} \binom{n}{j} p_a^j (1-p_a)^{n-j} = \mathbb{P}_{X \sim \mathrm{Binom}(n,p_a)}(X \geq h).$$

Accept if $p$-value $\leq \alpha$.

### E.2 Type I Error (False Positive)

**Theorem 11** (Per-Candidate Type I Error)**.** *Under the null hypothesis that $C$ has true precision $p_a$, the probability of materializing $C$ is $\leq \alpha$.*

*Proof.* Direct from definition of $p$-value. The test is exact (binomial, not asymptotic). □

**Theorem 12** (Family-Wise Error Rate (FWER))**.** *Let $M$ be the total number of candidates tested. Under global null (all candidates have precision $p_a$),*

$$\mathbb{P}(\textit{any false positive}) \leq M\alpha \quad \textit{(Bonferroni).}$$

*With $\alpha = 10^{-3}$, $M \sim 10^5$, Bonferroni gives FWER $\leq 0.1$. The actual FWER is lower due to dependence among candidates.*

### E.3 Type II Error (False Negative / Power)

**Theorem 13** (Detection Power)**.** *For a candidate with true precision $p > p_a$, coverage $n$, the power is*

$$\beta(n, p) = \mathbb{P}_{\text{Binom}(n,p)}(X \geq h_{crit})$$

*where $h_{crit} = \min\{h : \mathbb{P}_{\text{Binom}(n,p_a)}(X \geq h) \leq \alpha\}$.*

**Corollary 2** (Minimum Detectable Effect)**.** *For $n = c_{\min} = 4$, $p_a \approx 0.2$, $\alpha = 10^{-3}$: $h_{crit} = 4$ (need perfect precision). For $n = 10$, $h_{crit} = 7$ (70%). The filter requires* high precision on sufficient coverage *— this is by design to avoid overfitting sparse conjunctions.*

## F Rule Recovery Guarantees

### F.1 Setting

Let the true oracle contain a set of ground-truth compositional rules $\mathcal{R}^* = \{C_j^* \to a_j\}$ with arity $k_j \leq K$. Assume:

1. (Coverage) $\forall j$, $\text{cov}_{a_j}(C_j^*) \geq c_{\min}$ in the explore data.
2. (Precision) $\forall j$, $\text{prec}_{a_j}(C_j^*) \geq \theta + \delta$ for some $\delta > 0$.
3. (Separation) $\forall j$, $p$-value$(C_j^*) \leq \alpha/2$ with high probability.

**Theorem 14** (Recovery of Ground-Truth Rules)**.** *Under the above assumptions, with probability $\geq 1 - \eta$ (over data sampling), every $C_j^* \in \mathcal{R}^*$ is materialized as an ACTIVE rule.*

*Proof.* By Assumption 1, $C_j^*$ passes coverage pruning at all levels $\leq k_j$. By Assumption 2 and 3, it passes the binomial filter. The algorithm evaluates all candidates up to arity $K$, so $C_j^*$ is considered and materialized. □

**Remark 2.** *Phase 3 results: 4/6 oracle rules recovered in 100% of seeds (exact match); 2/6 recovered in 50-83%. The missed rules (cold $\wedge$ elder $\to$ obey, warm $\wedge$ stranger $\wedge$ spring $\wedge$ dawn $\to$ approach) violate Assumption 1 or 2 due to priority overlap with other rules.*

## G Phase Transition in Rule Recovery

Define the *effective branching factor* $b_{\text{eff}}(k) = |S_k^{\text{supported}}|/|S_{k-1}^{\text{supported}}|$.

**Proposition 1** (Phase Transition)**.** *If $b_{\mathit{eff}}(k) > 1$ for all $k \leq K$, the candidate set grows exponentially and induction is intractable. If $b_{\mathit{eff}}(k) < 1$ for some $k$, the search terminates early. The transition occurs at the critical coverage threshold $c_{\min}^*$ where $\mathbb{E}[b_{\mathit{eff}}(k)] = 1$.*

In TR-Core, $c_{\min} = 4$ places the system in the subcritical regime ($b_{\text{eff}} < 1$ for $k \geq 3$), enabling tractable discovery of arity-4 rules.

## H Comparison to Alternative Inducers

Table 4: Complexity comparison of induction methods

| Method | Time | Space | Statistical Guarantee |
|---|---|---|---|
| Apriori (this work) | $\mathcal{O}(NL + L^K)$ | $\mathcal{O}(N + L)$ | Exact binomial test |
| Decision Tree (CART) | $\mathcal{O}(NL \log N)$ | $\mathcal{O}(N)$ | Heuristic (impurity) |
| Decision List (greedy) | $\mathcal{O}(NL^K)$ | $\mathcal{O}(N)$ | Heuristic (lift/accuracy) |
| k-NN | $\mathcal{O}(NL)$ (query) | $\mathcal{O}(NL)$ | None (non-parametric) |
| Random Forest | $\mathcal{O}(TNL \log N)$ | $\mathcal{O}(TN)$ | Asymptotic consistency |

## I Conclusion

The Apriori induction engine in TR-Core has:

- **Worst-case time** $\mathcal{O}(L^K)$ — exponential in max arity, but $K = 4$, $L \leq 25$ makes it tractable.
- **Expected time** $\mathcal{O}((1 + \beta L)^K)$ — sub-exponential due to anti-monotonicity pruning with $c_{\min} = 4$.
- **Space** $\mathcal{O}(N + L)$ — linear in data, independent of candidate space.
- **Statistical guarantee** — exact binomial test controls per-candidate Type I error at $\alpha = 10^{-3}$; high precision threshold $\theta = 0.95$ ensures rules are behaviorally meaningful.
- **Recovery guarantee** — ground-truth compositional rules are recovered if they have sufficient coverage and precision in explore data.

The combinatorial explosion is tamed by the coverage floor $c_{\min}$, which induces a phase transition to a subcritical branching process. This is the key theoretical insight: *the induction engine is tractable precisely because the environment is sparse and the coverage threshold is set above the noise floor.*